

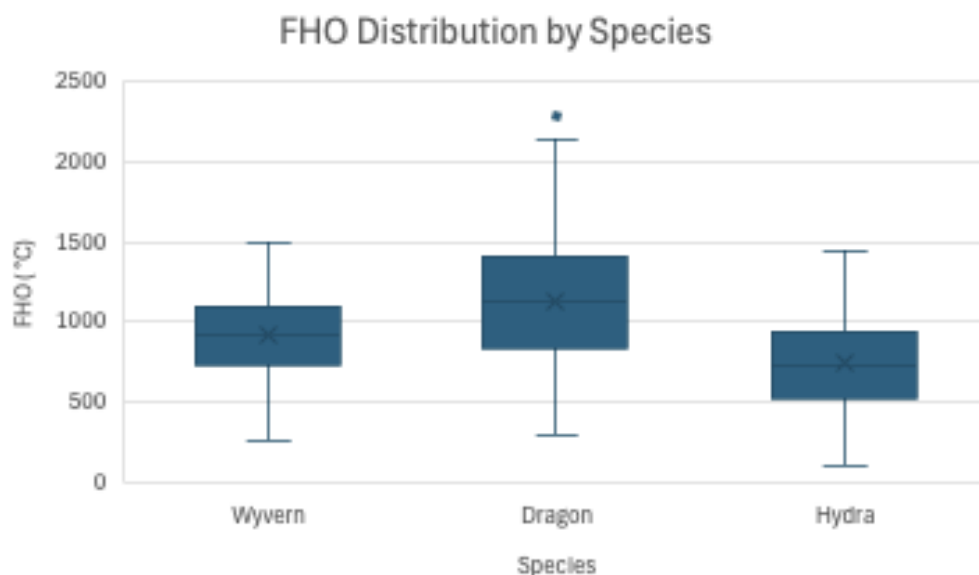
SPC:

Since species is a categorical variable with three levels (Dragon, Wyvern, Hydra), it can't be used directly in a regression formula. Therefore, it was encoded into two binary dummy variables: SPC_Dragon and SPC_Hydra, with Wyvern used as the reference category.

Fitted model: $FHO = 916.01 + 222.85 \times SPC_Dragon - 168.28 \times SPC_Hydra$

Training R ²	0.17220338
Test R ²	0.08906893
Intercept	916.006667
Slope: Hydra	-168.27834
Slope: Dragon	222.850476

Plot:



Interpretation:

The intercept of 916.01°C represents the predicted FHO for a Wyvern (the baseline species). The coefficient for Dragon (+222.85) means Dragons produce flames almost 223°C hotter than Wyverns. The coefficient for Hydra (−168.28) means Hydras burn approximately 168°C cooler than Wyverns. These differences imply Dragons are typically the most powerful species while Hydras appear to produce less intense individual flame output.

Performance:

The training R² is 0.1722, meaning species alone explains 17.22% of the variation in FHO across the training set. The low R² suggests that the majority of variation in flame output is driven by other factors not captured by species alone.

The test R^2 is 0.0891 which is a noticeable drop from the training R^2 of 0.1722. This suggests the model doesn't generalise strongly to unseen data.

Species is a statistically meaningful but weak predictor of FHO on its own. It captures differences between species groups but leaves the large variation within each species unexplained. It's more useful as one variable among several in a multiple regression model such as mass and wingspan.